

Recap: Location Privacy and Anonymous Communications

Anonymous Communications

1. Types of communication privacy protection

Communication privacy can protect several aspects: the sender of a message, the recipient of a message, the fact that two parties communicate (linkability), the author of a document, the identity of the server storing a document, the end user who received a document, and information about which documents are stored on a certain peer. The technical approaches to achieve this include onion routing, mixes, and proxies. (Slides 5, 9)

2. Main differences between a TOR network and a mixnet

TOR and mix networks both aim to provide anonymous communication, but they differ in key ways. TOR is a low-latency system that anonymizes TCP streams (not individual packets), uses fixed-size 512-byte cells, and routes data through a circuit of 3 randomly selected relays without delaying or reordering packets — it prioritizes performance with delays ideally under one second. Mix networks, by contrast, deliberately collect, delay, and reorder messages before forwarding them in random order, which provides stronger protection against a global observer who can watch all network links, but at the cost of higher latency. TOR relies on layered encryption (peeling layers at each hop) and establishes circuits using public-key crypto followed by symmetric-key crypto for data transfer. Mixes also use encryption but add the critical step of batching and shuffling messages to break the correlation between inputs and outputs. (Slides 10, 13, 15, 20, 25)

3. Protection achieved by a mix

A mix hides the correspondence between its incoming and outgoing messages. It collects multiple messages, then sends them out in a randomized order, so an observer watching both sides of the mix cannot determine which input message corresponds to which output message. When messages pass through a chain of mixes, even a single honest mix in the chain is sufficient to provide anonymity. Message padding (making all messages the same size) and re-encryption further prevent an attacker from correlating messages by size or content. (Slides 20, 25)

4. Attacks on a mix

The primary attack discussed is the *n-1 attack*, which is an active attack. The attacker floods a mix with $n-1$ of their own messages so that only one legitimate message is in the batch. When the mix fires, the attacker recognizes all of their own messages in the output and can therefore identify the single remaining message and its destination. This attack works most directly against threshold mixes, but a strong attacker could also delay legitimate messages toward a timed mix to achieve the same effect. The

pool mix was designed specifically to make this attack harder by always retaining a pool of old messages in the buffer. (Slides 23, 24)

Location Privacy

1. Mixed zones concept for location privacy

The mixed zone concept (by Beresford and Stajano, 2003) provides anonymity by dividing geographic space into “application zones” where users’ locations are tracked, and “mixed zones” where no location observations are made. Users carry pseudonyms that are changed whenever they enter or exit a mixed zone. When a user enters a mixed zone (ingress event), their location is no longer observed until they leave (egress event), at which point they receive a new pseudonym. This breaks the continuity of tracking — an observer cannot link the identity entering a mixed zone to the identity leaving it, analogous to how a network mix breaks the link between input and output messages. The design and configuration of these zones (size, placement, traffic volume) determines how much anonymity is actually provided. (Slide 33)

2. Threats from collecting user location

Collecting user location data poses several threats. An adversary can determine where a person lives, works, and travels, and can infer sensitive activities such as visits to medical facilities, political gatherings, or other private locations. In the vehicular context specifically, connected cars enable data collection and reporting, third-party apps in vehicles have broad API access, cameras can track the environment and driving behavior (potentially without GDPR-compliant consent), navigation data may be collected, and the car can access smartphone data. These threats span identity tracking, spatial tracking, and temporal tracking — knowing who someone is, where they are, and when they were there. (Slides 29, 30, 39)

3. Scenarios of protecting location versus protecting user identity

Location privacy involves three dimensions that can be protected independently or in combination: the user’s identity (ID), their spatial position, and the time of localization. Slide 30 presents a detailed table of scenarios. For example, protecting all three attributes means preventing anyone from inferring where a user lives from multiple traces. Protecting only identity (but revealing position and time) enables things like contributing anonymous GPS traces to mapping services. Protecting only position (but revealing identity and time) allows use cases like telling friends you’re in the inner city without revealing you’re specifically at a bar. Protecting only time means sharing a hiking trail with friends without revealing you’re currently away from home. The key insight is that different applications call for different combinations of protection. (Slide 30)

Vehicular Privacy

1. General privacy requirements for connected cars

Connected cars must ensure anonymity against unauthorized parties, making it impossible to link identifiers to real-world objects (person, device, or car). The driver should decide if and when personal data such as location, images, or driving behavior is disclosed. At the same time, the system must preserve *resolvable pseudonymity* — authorized parties (such as law enforcement in case of accidents or emergencies) should be able to resolve pseudonyms to real identities. This enables accountability and non-repudiation while still protecting the driver's privacy against unauthorized tracking. (Slide 42)

2. Vehicle pseudonym changing (purpose, design rules)

The purpose of pseudonym changing is to prevent tracking of vehicles over time. The ETSI security concept uses short-lived, anonymized “Authorization Tickets” (pseudonym certificates) for V2X communication that do not contain identity information. The design rules include: only temporary identifying information is allowed in external communications; all identifiers across all protocol layers (link to session) must be changed simultaneously (otherwise they remain linkable); pseudonyms should be changed periodically (e.g. every 120 seconds) and preferably on motorways away from home or work locations (to avoid linking pseudonyms to sensitive places); too-frequent changes should be avoided as they can disrupt message addressing and routing; and location accuracy should never be more precise than necessary. (Slides 43, 44)